

SimICU — Instructor Guide

C.A.R.E. — Cardiac Advanced Resuscitation Education · In-Situ Simulation Tool

This guide covers how to run the simulator, how to read the monitor, the worked scenarios below (the exact monitor values at each step and the teaching intent behind each move), how to build and edit your own scenarios, and how to import and export them between instructors. Scenario data is taken from the app's own source, so the numbers here match what the simulator displays. The CSU-ALS cases are scripted in their own document, issued separately.

Not a medical device. SimICU is a training aid. Clinical values are simplified for education and must not be used for patient care.

Sections marked  cover the **scenario builder**, which is part of the SimICU Full Access subscription. Everything else — the built-in scenarios, both device views, pairing, and the full instructor control panel — is free, with no account required.

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1. How the simulator works

The simulator runs on **two devices** that stay in sync over the local network:

Device	Role	What it shows
iPad / tablet	Monitor View	The full-screen bedside monitor the learners watch — live ECG, arterial line, PA catheter, SpO ₂ pleth, capnography, numerics, and alarms. Also holds the defibrillator / pacer controls.
iPhone / phone	Instructor View	Your hidden control panel — 5 tabs (Vitals, Rhythm, Actions, Scenario, Alarms). Nothing here is visible to learners.

Pairing (no server, no laptop): put both devices on the same Wi-Fi network (no internet needed — a classroom router or a Personal Hotspot is enough), open Monitor View on the tablet and Instructor View on the phone. The phone finds the tablet automatically. The tablet shows a **4-digit pairing code** in its top bar (tap the status dot to see it big) — enter that code on the phone when prompted and the two link. The code keeps anyone else on the Wi-Fi from controlling your session. When linked, every change you make appears on the tablet instantly; if the network drops, the two fall back to independent operation and re-link automatically. If an instructor's phone restarts mid-scenario, it re-pairs and **adopts the running session** — it won't reset the monitor.

The five instructor tabs:

- **Vitals** — set any number directly (HR, BP, SpO₂, PA, CVP, CO/CI/SVR, temp...), toggle which lines the learners see (Art, PA, ETCO₂, Hemodynamics), and switch the **IABP** balloon-pump augmentation on the arterial trace.
- **Rhythm** — pick the underlying ECG rhythm (see [Appendix A](#)).
- **Actions** — one-tap interventions (drugs, shocks, CPR, O₂...) that nudge the vitals (see [Appendix B](#)). You can also define custom actions.
- **Scenario** — the scripted cases in this guide. Run them, duplicate/edit them, or build your own.
- **Alarms** — set the alarm limits that drive the monitor's warning banner.

2. Running a scenario

1. In **Instructor View**, open the **Scenario** tab. You'll see the **CSU-ALS** folder (10 cases), the **Built-in** list (6 cases), and **My Scenarios** (anything you've created).
2. **Tap a scenario** to open its step list. Each row is one clinical state ("step" for built-ins, "slide" for CSU cases).
3. **Tap a step** to push that state to the monitor. The row is marked **LIVE** and the monitor updates immediately.
4. Advance by tapping the next step when *you* decide the learners are ready — **scenarios are instructor-paced, never on a timer**.

Steps are a decision tree, not a fixed reel. Especially in the CSU-ALS library, some slides are *correct-path* outcomes and others are *deterioration paths* that show what happens after a wrong move. You pick the slide that matches what the team actually does. Each scenario below includes a **branch map** so you know which slide to jump to.

Two operating habits worth forming:

- **Always start from the first step ("Initial Status").** Loading a step only changes the values that step specifies; anything it doesn't set carries over from whatever was on the monitor before. Starting at step 1 establishes a clean baseline so nothing lingers from a previous case.
- **Improvise freely.** The scripted slides are a backbone. At any time you can reach for the **Actions**, **Vitals**, or **Rhythm** tabs — or the monitor's defib/pacer — to respond to something the team does that the script didn't anticipate.

Make it your own: tap **Duplicate** on any built-in case to get an editable copy, or + **New Scenario** to build one from scratch (set the vitals you want on the other tabs, then capture each stage as a step). Scenarios can be exported and imported as JSON to share between instructors.

3. Reading the monitor

The learner's monitor can show up to five waveforms plus a numeric panel:

Trace	Colour	Driven by
ECG (lead II)	green	the selected rhythm + heart rate
ART (arterial line)	red	art systolic/diastolic; shows IABP 1:1 augmentation when the balloon pump is on
PA (pulmonary artery catheter)	yellow	PA systolic/diastolic (+ PCWP); can be hidden per-case
SpO₂ pleth	cyan	oxygen saturation; flatlines when there is no pulse
ETCO₂ capnography	white	end-tidal CO ₂ + respiratory rate

Numeric panel: HR, SpO₂, NIBP, ETCO₂, RR, PA/PCWP, CVP, temperature, and the hemodynamic set (CO, CI, SVR, SvO₂).

Perfusion honesty: in pulseless states (V-Fib, asystole, PEA) the SpO₂ and arterial traces go flat even if a number is shown — exactly as they would at the bedside. Some CSU cases deliberately **hide the PA line** ("CVP only") until a step where the PA catheter "starts working," which is itself a teaching cue.

4. Scenario walkthroughs

The six built-in cases, step by step. Each runs presentation → deterioration → arrest or recovery, and suits ACLS-style teaching and warm-ups. Vitals shown are the key values on the monitor at each step; the **Cue** column is the instructor note that appears in the app.

The **CSU-ALS library** in the Scenario tab runs exactly as described in §2; those ten cases are scripted in their own document, issued separately.

4.1 Inferior STEMI → Cardiogenic Shock

62 y Male · 82 kg · Chest pain + diaphoresis ×45 min

Teaching focus: recognising inferior infarction, watching a pump fail into cardiogenic shock (rising filling pressures, falling CO), and running the arrest.

#	Step	Monitor	Cue
1	Presentation	HR 88 · Inferior STEMI · SpO ₂ 94 · BP 90/60 · PA 42/22 (PCWP 24) · CO 3.2 / CI 1.8 · SVR 1600	EMS arrival. Inferior STEMI pattern, BP low-normal.
2	Hemodynamic Decline	HR 108 · Inferior STEMI · SpO ₂ 90 · BP 76/46 · PA 50/28 (PCWP 28) · CO 2.4 / CI 1.3 · SVR 1667	Pressure dropping, altered mentation. Consider vasopressors.
3	V-Fib Arrest	HR 0 · V-Fib · SpO ₂ 68 · BP 0/0	Cardiac arrest. Initiate ACLS, charge defibrillator.
4	ROSC	HR 48 · Sinus Brady · SpO ₂ 82 · BP 86/56 · RR 10	After 2 shocks + epinephrine. Weak pulse returned.

4.2 Septic Shock

58 y Female · 70 kg · 3-day fever, confusion, hypotension

Teaching focus: warm distributive shock (low SVR, high CO), and the response to the sepsis bundle — fluids, then vasopressors, then improvement.

#	Step	Monitor	Cue
1	Initial Assessment	HR 120 · Sinus Tach · SpO ₂ 93 · BP 84/46 · Temp 39.1 °C · SVR 514 · CO 8.2 / CI 5.4 · SvO ₂ 82	Warm distributive shock, high lactate. Broad-spectrum ABX ordered.
2	Post Fluid Bolus	HR 110 · SpO ₂ 95 · BP 96/56 · CO 7.4 / CI 4.8 · SVR 685	30 cc/kg crystalloid given. Partial response.
3	Vasopressor Titration	HR 100 · BP 110/66 · RR 22 · SVR 807	Norepinephrine 0.15 mcg/kg/min. MAP improving.
4	Clinical Improvement	HR 95 · SpO ₂ 97 · BP 118/72 · Temp 38.4 °C · CO 6.1 / CI 4.0 · SVR 1067	MAP >65, urine output improving, lactate trending down.

4.3 Cardiac Arrest / ROSC

55 y Male · 88 kg · Witnessed collapse — downtime ~2 min

Teaching focus: the full ACLS cycle through changing arrest rhythms — V-Fib → asystole → PEA → ROSC — including treating reversible causes (H's & T's).

#	Step	Monitor	Cue
1	V-Fib — Shock Now	HR 0 · V-Fib · SpO ₂ 72 · BP 0/0	Rhythm confirmed VF. Prepare defibrillator 360 J.
2	Post-Shock Asystole	HR 0 · Asystole	Continue CPR, give Epi 1 mg, 2-min cycle.
3	PEA — Continue CPR	HR 55 · PEA	Organized rhythm, no pulse. Treat reversible causes (H's & T's).
4	ROSC — Sinus Brady	HR 46 · Sinus Brady · SpO ₂ 80 · ART 82/50 · NIBP 88/58	Pulse present, weak sinus brady. Targeted temp management.
5	Stabilizing	HR 74 · NSR · SpO ₂ 96 · BP 118/74	BP improving, preparing for cath-lab transfer.

4.4 Hypertensive Emergency

52 y Male · 90 kg · Severe headache, vision changes, BP 218/132

Teaching focus: controlled BP reduction (~25 % MAP over the first hour), titrating labetalol then a nicardipine drip without overshooting into hypoperfusion.

#	Step	Monitor	Cue
1	Presentation	HR 94 · Anterior STEMI · SpO ₂ 96 · BP 218/132 · SVR 1843 · CO 6.8	BP critically elevated, GCS 14, anterior ST changes. Consider end-organ damage.
2	IV Labetalol given	HR 82 · BP 198/118 · SVR 1655	20 mg IV push. Monitor closely ×10 min.
3	Nicardipine gtt started	HR 78 · BP 176/104 · SVR 1459	5 mg/hr. Titrate to 25 % MAP reduction over 1 h.
4	Target BP achieved	HR 74 · NSR · BP 160/100 · SVR 1365	MAP reduced ~25 % from baseline. Watch for hypoperfusion, admit ICU.

4.5 Respiratory Failure → Arrest

68 y Female · 75 kg · Progressive dyspnea, SpO₂ 84 % on RA

Teaching focus: escalating hypoxic/hypercapnic respiratory failure, the rising ETCO₂ of CO₂ retention, and a hypoxic arrest with recovery after intubation.

#	Step	Monitor	Cue
1	Initial Presentation	HR 118 · Sinus Tach · SpO ₂ 84 · BP 148/88 · RR 28 · ETCO ₂ 22	Severe hypoxemia, tachypneic, accessory muscle use. High-flow O ₂ started.
2	Declining — BiPAP started	HR 126 · SpO ₂ 78 · RR 32 · ETCO ₂ 56 · BP 158/94	Worsening; SpO ₂ not improving, ETCO ₂ rising (CO ₂ retention).
3	Pre-Arrest	HR 138 · SpO ₂ 68 · RR 36 · ETCO ₂ 68 · BP 162/98	Altered mental status. Preparing for RSI intubation.
4	V-Fib Arrest	HR 0 · V-Fib · SpO ₂ 52 · BP 0/0 · ETCO ₂ 8	Hypoxic cardiac arrest. Initiate ACLS.
5	ROSC Post-Intubation	HR 88 · NSR · SpO ₂ 92 · BP 102/62 · RR 14 · ETCO ₂ 38	ROSC, intubated, ventilated. SpO ₂ improving.

4.6 Massive Pulmonary Embolism

47 y Female · 68 kg · Sudden pleuritic chest pain + dyspnea after 8 h flight

Teaching focus: obstructive shock from RV strain — high PA pressures with a falling CO — and the hemodynamic turnaround after systemic thrombolysis.

#	Step	Monitor	Cue
1	Presentation	HR 116 · Sinus Tach · SpO ₂ 88 · BP 96/58 · PA 58/28 (PCWP 8) · CO 3.0 / CI 2.0 · SVR 1724	RV strain, sinus tach, hypoxic. S1Q3T3 pattern.
2	Hemodynamic Collapse	HR 132 · SpO ₂ 82 · BP 78/44 · PA 68/34 (PCWP 6) · CO 2.0 / CI 1.3 · SVR 1973	RV failure, obstructive shock. Consider systemic thrombolytics.
3	tPA given — Monitoring	HR 118 · SpO ₂ 86 · BP 88/52 · PA 60/28 · CO 2.6	100 mg alteplase over 2 h. Watch for improvement or bleed.
4	Responding to Lytics	HR 104 · SpO ₂ 93 · BP 108/66 · PA 42/18 · CO 3.8 / CI 2.5 · SVR 1558	Pulmonary pressures improving, SpO ₂ rising, BP stabilising.

5. Building your own scenarios

 **Requires Full Access.** The scenario builder — creating, editing, duplicating, importing, and exporting — is part of the **SimICU Full Access** subscription (\$4.99/month or \$49.99/year, each starting with a free 3-day trial). Everything else in this guide is free forever: the six built-in scenarios, both device views, pairing, and the full instructor control panel. Actions that need a subscription are marked with a  in the app; tapping one opens the subscription screen, and you can return to the free scenario list at any time.

Everything below happens on the **Scenario** tab in Instructor View. Custom scenarios are saved locally on that device and are fully editable — unlike the Built-in and CSU-ALS libraries, which are read-only until you **Duplicate** them.

Starting from scratch

1. Tap **+ New Scenario** at the top of the Scenario list.
2. Fill in the **Scenario Name** and **Patient** fields (age, weight, sex, chief complaint) — these appear on the monitor's patient bar during the run.
3. The step list starts empty. To build your first step:
 - Go to **Vitals, Rhythm, and Actions** and set up the exact clinical state you want the learners to see first.
 - Come back to **Scenario** and tap **+ Capture current monitor as step**. This snapshots every value currently live on the monitor into a new step.
4. Rename the step (e.g. "Presentation") and add an **instructor note** — a short cue reminding you what's happening and what you're watching for. This note is for you; learners never see it.
5. Repeat: change vitals/rhythm again for the next clinical stage, come back, and **capture** again. Each capture only stores what's different from a blank state, but when *played*, a step applies on top of whatever the monitor already shows — so build your steps in the order you intend to run them.
6. Use the $\uparrow$ / $\downarrow$ buttons on a step to reorder it,  to delete it, or $\rightarrow$ **Recapture live vitals** to overwrite a step with the monitor's current values (handy if you tweak something after capturing).
7. Tap **✓ Save** when done. The scenario now appears under **My Scenarios**.

Starting from an existing case

You don't have to start blank. Open any **Built-in** or **CSU-ALS** scenario and tap **Duplicate** — this makes a fully editable copy (named "*original (copy)*") under My Scenarios, with all of its steps intact. This is the fastest way to build a variant of a stock case (a different patient demographic, an extra deterioration step, a modified drug dose) without recreating it from scratch.

Editing or deleting later

From **My Scenarios**, each card has **Edit** (reopens the same editor), **Export** (see [Import & export](#)), and **Delete** (with a confirmation — this can't be undone).

Tips for building good steps

- **Step 1 should be a stable, complete baseline.** Since a step only changes the fields it explicitly sets, and running always starts learners on step 1, make sure that first capture has every relevant value set the way you want — don't rely on carryover from whatever the monitor happened to show before.
 - **Write the instructor note like a stage direction, not a summary.** Good: "BP dropping, consider vasopressor — don't let them bolus fluids here." Less useful: "Hypotension."
 - **Keep steps to distinct clinical stages, not every vitals tweak.** 4–6 steps (presentation → decline → intervention → outcome) is typically enough for a focused 10–15 minute run.
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6. Import & export

 **Requires Full Access.** Both importing and exporting are part of the subscription described in §5.

Custom scenarios can be shared between instructors as JSON files, using each device's native Share sheet — no server or account involved.

Exporting

From **My Scenarios**, tap **Export** on any custom scenario (built-ins and CSU-ALS cases can be exported too, after **Duplicate**, or directly via the **Export** button in the scenario's Run view). This opens the iOS Share sheet with the scenario's JSON payload, titled `<scenario name>.medsim.json`. From there you can:

- **AirDrop** it directly to another instructor's device.
- **Save to Files** to keep it, back it up, or move it via iCloud Drive.
- **Message or email** it as an attachment.
- **Copy** the raw JSON text if you want to paste it somewhere (a notes app, a shared doc) for later import.

Importing

1. On the Scenario tab, tap **Import** (next to **+ New Scenario**).
2. Paste the scenario JSON into the prompt and confirm.
3. The app validates the JSON (it must at least have a `steps` array) and assigns it a new local ID, so importing never collides with or overwrites an existing scenario — even if you import the same file twice, you'll get two separate copies under **My Scenarios**.
4. Every imported step's vitals are run back through the same safety sanitizer used for live network sync before they're allowed to touch the monitor, so a corrupted or hand-edited JSON file can't push an invalid or out-of-range value to the display.

If the paste fails, you'll see "**Import failed — could not read scenario**"; double-check that you copied the complete JSON (starting at `{` and ending at the matching `}`).

Note: import/export moves the scenario script only — the patient demographics, step vitals, rhythms, and instructor notes. It does not touch the CSU-ALS passcode lock; imported scenarios always land in **My Scenarios**, never inside the CSU-ALS folder.

Appendix A — Rhythm library

Set on the **Rhythm** tab. These are the ECG morphologies the monitor can render.

Rhythm	Category	Description
Normal Sinus Rhythm	Normal	Regular 60–100 bpm, normal P-QRS-T
Sinus Tachycardia	Normal	Regular >100 bpm, sinus origin
Sinus Bradycardia	Normal	Regular <60 bpm, sinus origin
Atrial Fibrillation	Supraventricular	Irregularly irregular, no P waves, narrow QRS
Atrial Flutter (4:1)	Supraventricular	Sawtooth flutter ~300 bpm; QRS every 4th
SVT / AVNRT	Supraventricular	Regular narrow complex 150–220 bpm
Ventricular Tachycardia	Ventricular	Wide, monomorphic, 140–220 bpm
Ventricular Fibrillation	Ventricular	Chaotic baseline, no complexes
Pulseless Electrical Activity	Ventricular	Organized ECG, no pulse / no output
Asystole	Ventricular	Flat line
Idioventricular Rhythm	Ventricular	Wide escape, 20–40 bpm
1st Degree AV Block	Blocks	Prolonged PR >200 ms, all conducted
2nd Degree — Wenckebach	Blocks	Progressive PR lengthening, then dropped QRS
2nd Degree — Mobitz II	Blocks	Fixed PR, intermittent dropped QRS
3rd Degree (Complete) AV Block	Blocks	AV dissociation, independent P and QRS
Left Bundle Branch Block	Conduction	Wide QRS, broad notched R, inverted T
Right Bundle Branch Block	Conduction	Wide QRS, RSR' in V1, broad terminal S
Anterior STEMI	STEMI	ST elevation V1–V4, hyperacute T
Inferior STEMI	STEMI	ST elevation II/III/aVF
Ventricular Paced	Paced	Pacing spikes + wide LBBB-pattern QRS

Appendix B — Interventions & their effects

One-tap buttons on the **Actions** tab. Effects are deliberately simplified — they nudge the numbers so the monitor reacts believably to a treatment. Know what each does so it matches your narration:

Action	Effect on the monitor
Epinephrine 1 mg	HR +20 (cap 180), art +25/+10
Defibrillate 360 J	Only acts on VF/VT. ~55 % chance → NSR @ 52; otherwise → asystole. (<i>Probabilistic — for a guaranteed outcome, load the next scenario step instead.</i>)
Cardiovert 200 J	AF / SVT / VT → NSR @ 72
Start CPR	Arterial trace → 62/22 (compression pressure)
ROSC	→ Sinus Brady @ 48, art 85/52, NIBP 90/58, SpO ₂ +12
Atropine 0.5 mg	HR +15 (cap 200)
Amiodarone 300 mg	HR –15 (floor 30)
Adenosine 6 mg	SVT / AF → NSR @ 74
500 mL NS Bolus	NIBP +16/+9
Vasopressor Push	NIBP +22/+14, HR +6
High-Flow O ₂	SpO ₂ +8 (cap 100)
Intubate	SpO ₂ +6, RR → 14

You can also build **custom actions** on the Actions tab (choose a parameter, set an absolute value or a delta, give it an icon/colour).

Defib & pacer live on the **Monitor** screen — the team charges/shocks and turns on transcutaneous pacing there, and it drives the trace directly (useful for the VT and Bradycardia cases).

Appendix C — Facilitation tips

- **Pre-brief the line-up.** If a case is "CVP only," don't reveal the PA line early — the moment it "starts working" (e.g. Tamponade slide 4) is a teaching beat.
- **Let the wrong path play out.** The deterioration slides exist so a team can see the consequence of the wrong call. Show them, then reset to the earlier slide and let them try again.
- **Layer live changes over the script.** Between slides, use Actions/Vitals to reward good micro-decisions (e.g. a small SpO₂ bump after real O₂) without jumping the whole scenario.
- **Use the alarm limits.** The Alarms tab sets when the monitor's red banner fires — tighten them to force the team to notice a subtle trend, or loosen them to reduce noise for a novice group.
- **IABP cases.** Mitral Regurgitation and others end on balloon-pump support — flip the **IABP** toggle (Vitals tab) so the arterial trace shows the 1:1 diastolic augmentation the learners should recognise.

- **VT perfuses at whatever BP you set.** Unlike VF/asystole/PEA (always pulseless), ventricular tachycardia can be perfusing or pulseless in real life — so the sim leaves it to you: for **pulseless VT**, set the arterial BP to 0/0 and the trace flatlines. (The CSU-VT slide deliberately keeps 60/40 to show a barely-perfusing wide-complex tachycardia.)
 - **Arrest-state realism.** Drug/fluid buttons deliberately do nothing visible during VF/asystole/PEA — there's no circulation to respond. The press is still the team's action; you decide the response (e.g. tap a ROSC step).
 - **Build variants.** Duplicate a case, tweak a few numbers or add a slide, and save it under *My Scenarios*. Export as JSON to share with co-instructors.
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Generated from SimICU scenario data. Keep this guide with the app; update it if you add or edit scenarios.